

# Atanu Kundu

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## Research Area

📖 I am a researcher in formal verification and falsification of cyber-physical systems, with expertise in developing algorithms for bounded model checking and machine learning-based falsification.

- Cyber-Physical Systems
- Bounded Model Checking
- Falsification
- Machine Learning for Falsification
- AI-Assisted Safety Analysis

## Education

- |                                           |                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 – present<br><i>thesis submitted</i> | 📖 <b>PhD in Computer Science from IACS Kolkata, India.</b><br><b>Thesis title:</b> <i>Bounded Model Checking and Data-Driven Falsification for CPS.</i><br><b>Courses:</b> Big Data Analytics and Advanced Machine Learning <b>CGPA:</b> 8.50 |
| 2016 – 2018                               | 📖 <b>M.Sc. in Computer Science from Visva-Bharati, India.</b><br><b>Courses:</b> Advanced Data Structure, AI, DAA, etc <b>CGPA:</b> 7.93                                                                                                      |
| 2013 – 2016                               | 📖 <b>B.Sc. in Computer Science from The University of Burdwan.</b><br><b>Percentage:</b> 66.12                                                                                                                                                |

## Research Achievements

### List of Publications

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|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journal Papers<br><i>peer-reviewed</i>  | 📖 A. Kundu, S. Das, R. Ray, "SAT-Reach: A Bounded Model Checker for Affine hybrid Systems". ACM Trans. Embed. Compu. Syst vol. 22, no. 2, Jan. 2023, issn: 1539-9087. doi: 10.1145/3567425.                                                                                                                                                  |
|                                         | 📖 A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2025. doi: 10.1109/TCAD.2025.3608632.                                                                                                                                  |
| Int Conferences<br><i>peer-reviewed</i> | 📖 A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," in ISEC '24, ISBN: 9798400717673. doi: 10.1145/3641399.3641401                                                                                                                                                                                        |
|                                         | 📖 S. Gon, A. Kundu, and R. Ray, "Falsification of cyber-physical systems using causation-aware reinforcement learning," accepted in the 29th ACM International Conference on Hybrid Systems: Computation and Control, HSCC/ICCPS, 2026.                                                                                                      |
| Int Workshops                           | 📖 L. Bu, G. Frehse, A. Kundu, R. Ray, Y. Shi, and E. Zaffanella, "Arch-comp22 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in Proceedings of the 9th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH22), vol. 90, 2022, pp. 44–57. doi: 10.29007/lnzf |
|                                         | 📖 C. Menghi et al., "Arch-comp23 category report: Falsification," in Proceedings of the 10th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH23), G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 96, EasyChair, 2023, pp. 151–169. doi: 10.29007/6nqs.                             |

## Research Achievements (continued)

- T. Khandait et al., “Arch-comp 2024 category report: Falsification,” in Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems, G. Frehse and M. Althoff, Eds., ser. EPIc Series in Computing, vol. 103, EasyChair, 2024, pp. 122–144. doi: 10.29007/hgfv.
- L. Bu, A. Kundu, R. Ray, and Y. Shi, “Arch-comp24 category report: Hybrid systems with piecewise constant dynamics and bounded model checking,” in Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems, G. Frehse and M. Althoff, Eds., ser. EPIc Series in Computing, vol. 103, EasyChair, 2024, pp. 1–14. doi: 10.29007/nv67
- T. Khandait et al., “Arch-comp25 category report: Falsification,” in Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems, G. Frehse and M. Althoff, Eds., ser. EPIc Series in Computing, vol. 108, EasyChair, 2025, pp. 169–189. doi: 10.29007/dgnn.
- L. Bu, A. Kundu, R. Ray, and Y. Shi, “Arch-comp25 category report: Hybrid systems with piecewise constant dynamics and bounded model checking,” in Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems, G. Frehse and M. Althoff, Eds., ser. EPIc Series in Computing, vol. 108, EasyChair, 2025, pp. 1–14. doi: 10.29007/x3nv.

### List of Talks

Int Conference  
*Contributed  
Talks*

- Research paper **Falsification of cyber-physical systems using causation-aware reinforcement learning** presented at HSCC/ICCPS 2026, CPS-IoT 2026, Saint Malo, France.
- Research paper **Data-driven Falsification of Cyber-physical Systems** presented the paper at ISEC 2024, IIIT Bangalore.

Invited Talks

- Research paper **Data-driven Falsification of Cyber-physical Systems** published in IEEE TCAD 2025 and presented at RHPL co-located with FSTTCS 2025, BITS Pilani, GOA.
- A short presentation titled **A Framework for Detecting Unsafe Behavior in Cyber-Physical Systems** presented in Tech Symposium on Computing Trends 2023, organized by THALES associated with UNIVERSITY OF CALCUTTA.
- **SAT-Reach: A Bounded Model Checker for Affine Hybrid Systems** presented in the formal methods update meeting 2022 at IIT Delhi.

### External Achievements

Travel Grant

- Awarded **ACM SIGBED** travel grant for attending **CPS-IoT Week 2026**.
- Got an International Travel Support from IACS, Kolkata, India.

Certification

- **Special recognition for completing a short thesis presentation in exactly 5 minutes**. Awarded by THALES associated with UNIVERSITY OF CALCUTTA.

Fellowship

- Awarded Institute PhD Fellowship from IACS, Kolkata, India.

## Projects and Work Experiences

2017 - May 2018

- Raylay TCP over wireless network with free-space optical and worldwide interoperability for microwave access.  
*under the supervision of Mr. Subhasis Banerjee, Visva-Bharati, India.*

## Projects and Work Experiences (continued)

- May - Dec 2022     Building neural network models of the standard Cyber-Physical Systems (CPS) benchmarks.  
*Under the supervision of Dr. Rajarshi Ray, IACS Kolkata, India.*  
Building Feed-forward Neural Networks (FNNs) for CPS benchmarks such as hybrid automata and Simulink models. We built FNN models for systems with both linear and non-linear dynamics, as well as time-varying input signals. These models are approximations of the original models and can be useful in developing applications for such systems.

## Skills

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages         |  Native language Bengali. Fluent in English and Hindi.                                                                                                               |
| Coding            |  Proficient in C, C++, Python, and SMT-LIB2. Familiar with MATLAB and LaTeX.                                                                                         |
| Technical Domains |  Formal Verification, Hybrid Automata, Falsification, Reachability Analysis, SAT Solving, STL, Robustness Monitoring, and Surrogate Modeling using Machine Learning. |
| DevOps            |  Experience in git and Docker.                                                                                                                                       |
| Tools             |  FlexiFal, NNFal, XSpeed, SAT-Reach, SpaceEx, Flow*, dReach, and breach.                                                                                             |
| Soft Skill        |  Team Work, Time Management, Presentation Skills, and Leadership.                                                                                                    |
| Misc.             |  Academic research, teaching, training, consultation, LaTeX typesetting, and publishing.                                                                             |

## Teaching Experiences

- Spring 2022     Teaching assistant on Artificial Intelligence (COM 4211)  
Responsibilities included setting and evaluating lab assignments. Supervising students throughout the course to succeed in the lab assignments.
- Autumn 2023     Teaching assistance on Object-Oriented Programming with C++ Lab (COM 4111)  
My duties involved creating and assessing assignments and surprise tests. Additionally, I was responsible for overseeing students' progress during the course, ensuring their success in laboratory assignments.

## References

### Dr. Rajarshi Ray

Associate Professor

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### Dr. Ansuman Banerjee

Professor

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Indian Statistical Institute, Kolkata, India

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### Dr. Partha Basuchowdhuri

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